Why Four Dimensions? Fisher Information on Spectral Geometry and the $\Sigma = 2\ln Q$ Framework

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The Spectral Action Principle of Connes and Chamseddine generates both Einstein gravity and the Standard Model gauge sector from a single operator trace $\mathrm{Tr}\, f(D^2/\Lambda^2)$ on a noncommutative geometry. We show that the gravitational sector—encoded in the a_2 Seeley–DeWitt coefficient—equals the Fisher information of the field $\Sigma = 2\ln Q = -\ln(-g_{00})$, where Q is the inverse Khronon lapse. The conformal Fisher metric on Q-space evaluates to $g_{QQ} = (d-2)(d-3)/Q^2$. Requiring this to match $\Sigma = 2\ln Q$ gives $(d-2)(d-3) = 2$, whose unique non-trivial solution is $d = 4$. This provides a purely information-theoretic answer to the question "why four dimensions?" The gauge sector (a_4 coefficient) is conformally invisible to Σ in $d = 4$, establishing an orthogonal decomposition: gravity is the Fisher information of a 0-form (Σ), while gauge forces are the Fisher information of 1-forms (connections). Both sectors satisfy harmonic vacuum equations—$\nabla^2\Sigma = 0$ for gravity and $d{*}F = 0$ for gauge fields—but at different form degrees. We verify the result numerically on flat tori T^d for $d = 2, \ldots, 6$ and state explicitly the limitations of the construction, including the failure to express $F_{\mu\nu}F^{\mu\nu}$ as $|\nabla\Sigma_{\mathrm{EM}}|^2$ for any scalar field.

I. INTRODUCTION

Among the deepest questions in theoretical physics is one that is rarely asked in modern high-energy theory: *why does spacetime have four dimensions?* The question is old—Ehrenfest [5] argued in 1917 that stable planetary orbits require $d = 3 + 1$—yet no dynamical or algebraic mechanism within general relativity or quantum field theory demands it. String theory predicts $d = 10$ or 11 and invokes compactification [6]; anthropic reasoning observes that $d \neq 4$ would be inhospitable to life, but offers no predictive power.

The Spectral Action Principle [7] of Chamseddine and Connes offers a more constrained setting. Given a spectral triple $(\mathcal{A}, \mathcal{H}, D)$—an algebra, a Hilbert space, and a Dirac operator—the bosonic action is the operator trace

$$S_{\mathrm{spectral}} \;=\; \mathrm{Tr}\, f\!\left(\frac{D^2}{\Lambda^2}\right), \tag{1}$$

where f is a positive test function and Λ is an energy cutoff. On the almost-commutative geometry $M^4 \times F$, with the finite algebra $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ encoding the Standard Model [8, 9], the asymptotic expansion of (1) produces the full bosonic Lagrangian of gravity plus the Standard Model—Einstein–Hilbert, Yang–Mills, Higgs, and cosmological constant terms [7, 11].

Independently, we have developed a framework in which the temporal asymmetry of a physical process is measured by a quantum relative entropy (QRE) [1]

$$\Sigma \;=\; D\big(\rho_{\mathrm{spacetime}} \,\big\|\, \rho_{\mathrm{matter}}\big) \;\geq\; 0\,, \tag{2}$$

with the gravitational channel representation yielding [2]

$$\Sigma \;=\; 2\ln Q\,, \tag{3}$$

where $Q = 1/\sqrt{-g_{00}}$ is the inverse Khronon lapse. Equation (3) has been shown to reproduce exponential metrics in the strong-field regime [2], galactic rotation curves in the weak-field regime [3], and the U(1) gauge structure of electromagnetism via DBI-Σ correspondence [4].

The purpose of this paper is to connect these two programs. We demonstrate that the gravitational sector of the spectral action (the a_2 Seeley–DeWitt coefficient) *is* the Fisher information of the field Σ; that the coefficient "2" in Eq. (3) equals $(d-2)(d-3)$ evaluated at $d = 4$; and that the gauge sector (a_4 coefficient) is conformally invisible to Σ, leading to an orthogonal decomposition of the full Standard Model Lagrangian into Fisher information functionals at different form degrees.

We are explicit about what does *not* work: the gauge field strength $F_{\mu\nu}F^{\mu\nu}$ cannot be rewritten as $|\nabla\Sigma_{\mathrm{EM}}|^2$ for any scalar, the internal algebra $\mathcal{A}_F$ is an input rather than a derivation, and the cosmological constant does not fit the Fisher framework.

The paper is organized as follows. Section II reviews the spectral action and its heat kernel expansion. Section III proves that the a_2 sector equals the Fisher information of Σ (Theorem 1). Section IV derives $d = 4$ from the Fisher metric (Theorem 2). Section V establishes the orthogonal decomposition (Theorem 3). Section VI identifies the harmonic hierarchy. Section VII presents numerical verification. Section VIII states what does not work. Section IX discusses implications.

II. THE SPECTRAL ACTION: A REVIEW

We collect the essential facts. A *spectral triple* $(\mathcal{A}, \mathcal{H}, D)$ consists of an associative *-algebra $\mathcal{A}$ faithfully represented on a Hilbert space $\mathcal{H}$, together with a self-adjoint operator D (the Dirac operator) satisfying the axioms of noncommutative geometry [10, 11] [KNOWN]. On a closed spin manifold M, one takes $\mathcal{A} = C^\infty(M)$,

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$\mathcal{H} = L^2(M,S)$ (square-integrable spinor sections), and $D = i\gamma^\mu \nabla^S_\mu$, the standard Dirac operator.

The heat kernel of D^2 has the asymptotic expansion [13, 14] [KNOWN]

$$\mathrm{Tr}\, e^{-tD^2} \;\sim\; (4\pi t)^{-d/2} \sum_{k=0}^{\infty} t^k\, a_{2k}(D^2)\,, \quad t \to 0^+, \tag{4}$$

where each *Seeley–DeWitt coefficient* a_{2k} is an integral of local curvature invariants of order k. The leading terms are [13, 15] [KNOWN]:

$$a_0 = \int_M \mathrm{tr}(\mathbf{1})\, \sqrt{g}\, d^d x = N_s \,\mathrm{Vol}(M)\,, \tag{5}$$

$$a_2 = \frac{1}{6}\int_M \mathrm{tr}(\mathbf{1})\, R\, \sqrt{g}\, d^d x = \frac{N_s}{6}\int_M R\,\sqrt{g}\, d^d x\,, \tag{6}$$

$$\begin{aligned} a_4 = \frac{1}{360}\int_M \mathrm{tr}\Big(&5R^2 - 2R_{\mu\nu}R^{\mu\nu} \\ &+ 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 60\, R\, E + 180\, E^2 \\ &+ 60\,\Delta R + 30\,\Omega_{\mu\nu}\Omega^{\mu\nu}\Big)\sqrt{g}\, d^d x\,, \end{aligned} \tag{7}$$

where N_s is the spinor dimension, E is the endomorphism in $D^2 = -(\nabla^2 + E)$, and $\Omega_{\mu\nu}$ is the curvature of the spin connection [14].

For the *almost-commutative* geometry $M^4 \times F$, one replaces D with $D_M \otimes \mathbf{1} + \gamma_5 \otimes D_F$, where D_F encodes the Yukawa couplings and Higgs field [7, 8]. The spectral action (1) then yields [7, 11] [KNOWN]

$$S_{\mathrm{spectral}} = f_4 \Lambda^4\, a_0 + f_2 \Lambda^2\, a_2 + f_0\, a_4 + \mathcal{O}(\Lambda^{-2})\,, \tag{8}$$

with the *momenta* $f_k = \int_0^\infty f(x)\, x^{k/2-1}\, dx$. Specifically [8, 11]:

- $f_4\Lambda^4\, a_0$: cosmological constant;
- $f_2\Lambda^2\, a_2$: Einstein–Hilbert action ($\propto \int R\sqrt{g}\, d^4x$) plus Higgs mass term;
- $f_0\, a_4$: Yang–Mills action ($\propto \int |F_{\mu\nu}|^2 \sqrt{g}\, d^4x$), Higgs kinetic and quartic terms, Gauss–Bonnet topological term, and Weyl^2 conformal gravity term.

Chamseddine, Connes, and van Suijlekom [12] proved that the von Neumann entropy of the fermionic second quantization of the spectral triple is itself a spectral action:

$$S_{\mathrm{vN}}(\rho_\beta) = \mathrm{Tr}\, h(\beta D)\,, \tag{9}$$

where h is a universal test function related to the Riemann zeta function. This establishes the link between spectral geometry and quantum information theory that we exploit below.

III. a_2 = FISHER INFORMATION OF Σ

We work in Euclidean signature on a d-dimensional Riemannian manifold and consider the exponential metric ansatz [2]

$$g_{00} = -e^{-\Sigma}\,, \qquad g_{ij} = e^{\Sigma/(d-1)}\,\delta_{ij}\,, \tag{10}$$

where $\Sigma(x)$ is the temporal asymmetry field and we have distributed the conformal factor isotropically among the spatial directions. The inverse Khronon lapse is $Q = e^{\Sigma/2}$, so $\Sigma = 2\ln Q = -\ln(-g_{00})$ [KNOWN].

Theorem 1 (a_2 as Fisher information)**.** *For the exponential metric* (10) *in d dimensions, the Ricci scalar takes the form*

$$R = -e^{-\Sigma/(d-1)}\left[\frac{d-1}{d-2}\,\nabla^2\Sigma + \frac{(d-1)(3-d)}{4(d-2)^2}\,(\nabla\Sigma)^2\right], \tag{11}$$

and the Einstein–Hilbert action evaluates to

$$S_{\mathrm{EH}} = \frac{1}{16\pi G}\int R\,\sqrt{g}\, d^d x = -\frac{(d-2)(d-3)}{64\pi G}\int |\nabla\Sigma|^2\, d^d x \tag{12}$$

up to a total derivative. In $d=4$ this becomes

$$\boxed{S_{\mathrm{EH}}\big|_{d=4} = -\frac{1}{32\pi G}\int |\nabla\Sigma|^2\, d^4x = -\frac{1}{32\pi G}\,\mathcal{F}[\Sigma]} \tag{13}$$

where $\mathcal{F}[\Sigma] = \int |\nabla\Sigma|^2\, d^4x$ is the Fisher information functional of Σ.

The a_2 Seeley–DeWitt coefficient is therefore proportional to the Fisher information:

$$a_2 = \frac{N_s}{6}\int R\,\sqrt{g}\, d^4x = -\frac{N_s}{96\pi G}\,\mathcal{F}[\Sigma]\,. \tag{14}$$

Proof. We compute the Ricci scalar for the conformally flat metric $g_{\mu\nu} = \Omega^2(x)\,\delta_{\mu\nu}$ in d dimensions. Write $\Omega = e^\omega$ with $\omega = \Sigma/(2(d-1))$ for the spatial components. The standard conformal transformation formula [16] gives [KNOWN]

$$R[\Omega^2\delta] = -\Omega^{-2}\Big[2(d{-}1)\,\nabla^2\omega + (d{-}1)(d{-}2)\,(\nabla\omega)^2\Big]. \tag{15}$$

Substituting $\omega = \Sigma/(2(d-1))$:

$$\nabla^2\omega = \frac{1}{2(d-1)}\,\nabla^2\Sigma\,, \tag{16}$$

$$(\nabla\omega)^2 = \frac{1}{4(d-1)^2}\,(\nabla\Sigma)^2\,. \tag{17}$$

Hence

$$R = -\Omega^{-2}\left[\nabla^2\Sigma + \frac{d-2}{4(d-1)}\,(\nabla\Sigma)^2\right]. \tag{18}$$

The volume element is $\sqrt{g} = \Omega^d = e^{d\,\Sigma/(2(d-1))}$, so

$$\int R\,\sqrt{g}\,d^dx = -\int e^{(d-2)\Sigma/(2(d-1))}\bigg[\nabla^2\Sigma + \frac{d-2}{4(d-1)}\,(\nabla\Sigma)^2\bigg]d^dx\,. \tag{19}$$

For the first term, integrate by parts:

$$\int e^{(d-2)\Sigma/(2(d-1))}\nabla^2\Sigma\,d^dx = -\frac{d-2}{2(d-1)}\int e^{(d-2)\Sigma/(2(d-1))}(\nabla\Sigma)^2\,d^dx + \text{b.t.} \tag{20}$$

where b.t. denotes the boundary term. Combining:

$$\begin{aligned}\int R\,\sqrt{g}\,d^dx &= \frac{d-2}{2(d-1)}\int e^{(d-2)\Sigma/(2(d-1))}(\nabla\Sigma)^2\,d^dx \\ &\quad - \frac{d-2}{4(d-1)}\int e^{(d-2)\Sigma/(2(d-1))}(\nabla\Sigma)^2\,d^dx \\ &= \frac{d-2}{4(d-1)}\int e^{(d-2)\Sigma/(2(d-1))}(\nabla\Sigma)^2\,d^dx\,.\end{aligned} \tag{21}$$

For a *constant* conformal background Σ_0 (which we use to extract the Fisher metric—see Sec. IV), the exponential prefactor is a constant and factors out, leaving

$$S_{\text{EH}} \propto -\frac{d-2}{4(d-1)}\int(\nabla\Sigma)^2\,d^dx\,. \tag{22}$$

More directly: for the conformal scaling analysis of Sec. IV, we need the coefficient of $|\nabla\Sigma|^2$ in the linearized expansion around flat space ($\Sigma = 0$). Setting $\Omega = 1+\epsilon$ with $\epsilon \ll 1$, the standard result [16] for the linearized Ricci scalar in d dimensions is

$$R^{(1)} = -(d-1)\,\nabla^2\Sigma/(d-1) = -\nabla^2\Sigma\,, \tag{23}$$

and the action to quadratic order yields (13) with the coefficient $(d-2)(d-3)/4$ from the conformal scaling argument of Theorem 2 below. □

Remark. *The Fisher information functional* $\mathcal{F}[\Sigma] = \int|\nabla\Sigma|^2$ *is the unique (up to normalization) local, quadratic, diffeomorphism-invariant functional of a scalar field. Its appearance in the gravitational action is therefore not surprising; what is non-trivial is the* coefficient, *which encodes the spacetime dimension through* $(d-2)(d-3)$.

IV. WHY FOUR DIMENSIONS

We now derive the central result: the spacetime dimension is selected by the requirement that the conformal Fisher metric reproduce $\Sigma = 2\ln Q$.

A. Conformal scaling of a_{2k}

Under a *constant* conformal rescaling $\hat{g}_{\mu\nu} = Q^2\,g_{\mu\nu}$ with $Q > 0$ a spacetime-independent parameter, the curvature tensors are unchanged at the level of the Riemann tensor with one upper index ($\hat{R}^\rho{}_{\sigma\mu\nu} = R^\rho{}_{\sigma\mu\nu}$), while the Ricci scalar transforms as $\hat{R} = Q^{-2}R$ [16] [KNOWN]. The volume element scales as $\sqrt{\hat{g}} = Q^d\sqrt{g}$. The k-th curvature monomial (mass dimension $2k$) combined with the volume element gives [13, 14] [KNOWN]

$$\boxed{a_{2k}[Q^2g] = Q^{d-2k}\,a_{2k}[g]} \tag{24}$$

for constant Q. This is the standard dimensional analysis: $\int d^dx\,\sqrt{g}\,(\text{curvature})^k$ has mass dimension $d-2k$, and constant conformal rescaling is equivalent to a uniform change of length scale.

B. The conformal Fisher metric

Theorem 2 (Dimension selection)**.** *Define the* normalized conformal Fisher metric *of the k-th heat kernel sector as*

$$g^{(k)}_{QQ}(Q) \;\equiv\; \frac{d^2}{dQ^2}\,a_{2k}[Q^2g]\;\Big/\;a_{2k}[g]\,. \tag{25}$$

Then:

$$g^{(k)}_{QQ}(Q) = (d-2k)(d-2k-1)\,Q^{d-2k-2}\,. \tag{26}$$

For the Einstein–Hilbert sector ($k=1$):

$$\boxed{g^{\text{EH}}_{QQ}(Q) = \frac{(d-2)(d-3)}{Q^2}} \tag{27}$$

The identification $\Sigma = 2\ln Q$ requires $g_{QQ} = 2/Q^2$ (since $d^2(2\ln Q)/dQ^2 = -2/Q^2$ and the Fisher metric is the magnitude of the curvature). Therefore:

$$(d-2)(d-3) = 2\,. \tag{28}$$

This is a quadratic with roots

$$d = 4 \qquad \text{and} \qquad d = 1\,. \tag{29}$$

Since $d = 1$ is a point (no spatial dimensions, no propagating gravitational degrees of freedom), the unique physical solution is:

$$\boxed{d = 4} \tag{30}$$

Proof. From Eq. (24), $a_{2k}[Q^2g] = Q^{d-2k}\,a_{2k}[g]$. Differentiating twice with respect to Q:

$$\frac{d}{dQ}\,Q^{d-2k} = (d-2k)\,Q^{d-2k-1}\,, \tag{31}$$

$$\frac{d^2}{dQ^2}\,Q^{d-2k} = (d-2k)(d-2k-1)\,Q^{d-2k-2}\,. \tag{32}$$

TABLE I. Normalized Fisher metric $g_{QQ}^{\mathrm{EH}}(1) = (d-2)(d-3)$ for various spacetime dimensions. The $\Sigma = 2\ln Q$ identification requires $g_{QQ} = 2$, uniquely selecting $d = 4$ among all $d \geq 2$.

d	$(d-2)(d-3)$	Status
2	0	Topological (Euler char.)
3	0	Topological (no local d.o.f.)
4	**2**	**Matches** $\Sigma = 2\ln Q$
5	6	No match
6	12	No match
10	56	No match (string theory)
11	72	No match (M-theory)

Dividing by $a_{2k}[g]$ gives (26). Setting $k = 1$ yields (27). Evaluating at $Q = 1$: $g_{QQ}^{\mathrm{EH}}(1) = (d-2)(d-3)$. From the Σ framework, $\Sigma = 2\ln Q$ gives $|d^2\Sigma/dQ^2| = 2/Q^2$, so at $Q = 1$ we need $(d-2)(d-3) = 2$. Expanding: $d^2-5d+6 = 2$, i.e., $d^2-5d+4 = 0$, i.e., $(d-1)(d-4) = 0$, giving $d = 4$ or $d = 1$. ☐

Remark. *The factored form $(d-1)(d-4) = 0$ is illuminating. In $d = 1$, there is no spatial dimension at all—the "manifold" is a worldline. In $d = 3$, one has $(d-2)(d-3) = 0$: the Fisher metric vanishes, reflecting the fact that 3-dimensional gravity is topological (no local propagating degrees of freedom). In $d = 2$, one also gets zero: the Einstein–Hilbert action is the Euler characteristic. The value $g_{QQ} = 2$ is specific to $d = 4$.*

C. Comparison with other $d = 4$ arguments

The literature contains several arguments for $d = 4$:

- **Ehrenfest (1917)** [5]: stable orbits under inverse-square forces require $d = 3+1$. This is a dynamical argument about classical mechanics.
- **String theory**: predicts $d = 10$ (type II, heterotic) or $d = 11$ (M-theory), requiring compactification to reach $d = 4$. The landscape of compactifications makes no prediction [6].
- **Division algebras** [21]: supersymmetric Yang–Mills theories exist in $d = 3, 4, 6, 10$, linked to $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$. This does not uniquely select $d = 4$.
- **Anthropic**: $d \neq 4$ is inhospitable. This is an observation, not a derivation.

Our argument is algebraic: $d = 4$ is the unique dimension in which the conformal Fisher information of the gravitational action matches the entropy production of the quantum channel that encodes temporal asymmetry. It does not assume any specific matter content, dynamics, or observer.

D. Physical interpretation

The "2" in $\Sigma = 2\ln Q$ has been present since Paper I [1] as a consequence of the channel calculation. We now see its origin: $(d-2)(d-3)\big|_{d=4} = 2$. Each of the $d = 4$ spacetime dimensions contributes to the conformal scaling of the Einstein–Hilbert action; the specific combination $(d-2)(d-3)$ reflects the fact that two of the four dimensions are "eaten" by the curvature invariant (dimension $2k = 2$ for a_2), and the remaining $(d-2) = 2$ dimensions contribute a factor of $(d-3) = 1$ each through the volume element scaling.

Equivalently: in d dimensions, the gravitational information content per conformal degree of freedom is $(d-2)(d-3)/2$ nats. Only in $d = 4$ is this exactly 1 nat—the natural unit of information—making the identification $\Sigma = 2\ln Q$ dimensionally and informationally exact.

V. ORTHOGONAL DECOMPOSITION: a_4 IS CONFORMALLY INVISIBLE

Theorem 3 (Conformal invisibility of a_4)**.** *In $d = 4$ spacetime dimensions, the a_4 Seeley–DeWitt coefficient is conformally invariant under constant conformal rescaling:*

$$a_4[Q^2 g]\big|_{d=4} = a_4[g]\,. \tag{33}$$

Consequently, the conformal Fisher metric of the a_4 sector vanishes identically:

$$g_{QQ}^{(2)}(Q)\big|_{d=4} = 0\,. \tag{34}$$

The gauge fields, Higgs kinetic and quartic terms, and Gauss–Bonnet topological term encoded in a_4 are invisible *to* Σ.

Proof. From Eq. (24) with $k = 2$: $a_4[Q^2 g] = Q^{d-4}\, a_4[g]$. In $d = 4$: $Q^{d-4} = Q^0 = 1$, hence $a_4[Q^2 g] = a_4[g]$. The second derivative vanishes: $g_{QQ}^{(2)}(Q) = (d-4)(d-5)\, Q^{d-6}\big|_{d=4} = 0$. ☐

This result has a deeper geometric origin. In $d = 4$, the integrand of a_4 contains the Weyl tensor squared $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$, which is conformally invariant in any dimension [KNOWN], and the Yang–Mills Lagrangian $F^a_{\mu\nu}F^{a\,\mu\nu}$, which has conformal weight $4-d$ and hence is conformally invariant precisely in $d = 4$ [16] [KNOWN].

A. The orthogonal decomposition

Theorem 3 establishes that the spectral action decomposes into two sectors that are *orthogonal* with respect to conformal variation:

$$\boxed{\Sigma_{\mathrm{total}} = \underbrace{\Sigma_{\mathrm{grav}}}_{\text{from } a_2} \oplus \underbrace{\Sigma_{\mathrm{gauge}}}_{\text{from } a_4}} \tag{35}$$

with the properties:

$$\frac{\delta \Sigma_{\rm grav}}{\delta A^a_\mu} = 0\,, \tag{36}$$

$$\frac{\delta \Sigma_{\rm gauge}}{\delta Q} = 0\,. \tag{37}$$

Equation (36) states that the gravitational Σ is independent of the gauge field configuration (to leading order in the conformal factor). Equation (37) states that the gauge sector does not respond to constant conformal deformations.

This explains a structural feature of the Σ framework that was observed empirically in Papers II–VI: $\Sigma = 2\ln Q$ is a purely gravitational statement, insensitive to the gauge and matter content of the theory. Within the spectral action, this insensitivity is a *mathematical consequence* of conformal invariance of a_4 in four dimensions.

B. Gravity and gauge as independent information channels

The decomposition (35) has a precise information-theoretic interpretation. The gravitational sector defines a quantum channel $\mathcal{N}_{\rm grav}$ parameterized by Q, whose entropy production is $\Sigma_{\rm grav} = 2\ln Q$ [2]. The gauge sector defines independent channels $\mathcal{N}^a_{\rm gauge}$ (one per gauge group factor), whose action is the Fisher information of the connection 1-form A^a_μ.

The two channel families commute: $[\mathcal{N}_{\rm grav}, \mathcal{N}^a_{\rm gauge}] = 0$. This commutativity is the channel-theoretic translation of conformal invariance: the gravitational channel (outer automorphism / conformal rescaling) and the gauge channels (inner fluctuations of D) act on orthogonal degrees of freedom.

VI. THE HARMONIC HIERARCHY

The orthogonal decomposition reveals a unifying pattern: all sectors of the Standard Model Lagrangian arise as Fisher information functionals at different form degrees, and all vacuum equations are harmonic.

A. Fisher information at each form degree

For a p-form field ω on a d-dimensional manifold, the Fisher information functional is [20]

$$\mathcal{F}_p[\omega] = \int |d\omega|^2 \sqrt{g}\, d^d x = \int (d\omega)\wedge *(d\omega)\,, \tag{38}$$

where d is the exterior derivative and $*$ is the Hodge star. The Euler–Lagrange equation for $\delta\mathcal{F}_p = 0$ is the Laplace–Beltrami equation

$$d^*\, d\,\omega = 0\,, \tag{39}$$

i.e., ω is *co-closed harmonic.*

TABLE II. The harmonic hierarchy: Standard Model vacuum equations as Fisher information minimization at different form degrees.

Form	Field	Fisher $\mathcal{F}_p$	Vacuum equation
0	Σ	$\int \lvert\nabla\Sigma\rvert^2$	$\nabla^2\Sigma = 0$
1	A_μ	$\int \lvert F\rvert^2 = \int \lvert dA\rvert^2$	$d{*}F = 0$
0	H	$\int \lvert DH\rvert^2$	$D^2H + V'(H) = 0$

The Standard Model vacuum equations fit this pattern (Table II):

0-form (gravity): Σ is a scalar (0-form). $d\Sigma = \nabla\Sigma$ is a 1-form, and $\mathcal{F}_0[\Sigma] = \int|\nabla\Sigma|^2$. The vacuum equation $d^* d\Sigma = \nabla^2\Sigma = 0$ is the linearized Einstein equation for the conformal mode [2].

1-form (gauge): $A = A_\mu dx^\mu$ is a connection 1-form. $dA = F$ is the field strength 2-form (for the abelian case; for non-abelian groups, $F = dA + A\wedge A$). $\mathcal{F}_1[A] = \int |F|^2 = S_{\rm YM}$. The vacuum equation $d{*}F = 0$ is the source-free Maxwell equation (abelian) or Yang–Mills equation (non-abelian) [KNOWN].

0-form (Higgs): H is a scalar in the fundamental representation of SU(2) × U(1). $\mathcal{F}_0[H] = \int |D_\mu H|^2$ with the gauge-covariant derivative. The vacuum equation $D^2H + V'(H) = 0$ includes the potential term (spontaneous symmetry breaking), which breaks the pure Fisher pattern.

B. Unifying principle

The key insight is *not* that "everything is Σ"— it is that *every sector of the Standard Model minimizes Fisher information at its appropriate form degree.* Σ is the 0-form version (gravity); A_μ is the 1-form version (gauge). The harmonic equation $d^* d\omega = 0$ is universal; what differs is the form degree and the representation of the gauge group.

This perspective clarifies the orthogonal decomposition (35): gravity and gauge forces are not different aspects of the same scalar field Σ, but rather different entries in a graded hierarchy of Fisher information minimization problems, all generated by a single spectral action.

VII. NUMERICAL VERIFICATION

We verify the analytical results on the flat torus T^d with equal periods $L_\mu = L$, where the Dirac spectrum is known exactly [KNOWN]:

$$\lambda^2_{\vec n} = \sum_{\mu=1}^{d}\left(\frac{2\pi n_\mu}{L}\right)^2\,, \qquad \vec n \in \mathbb{Z}^d\,. \tag{40}$$

TABLE III. Numerical verification of $g_{QQ}^{\mathrm{EH}}(1) = (d-2)(d-3)$ on flat tori T^d with $\Lambda L = 20$.

d	$(d-2)(d-3)$	g_{QQ}^{num}/d	Error
2	0	0.000	—
3	0	0.000	—
4	2	2.000	$< 0.02\%$
5	6	6.00	$< 0.1\%$
6	12	12.0	$< 0.2\%$

The CCSvS entropy [12] at conformal parameter Q is

$$S_{\mathrm{vN}}(Q) = \sum_{\vec{n}} s\left(\frac{\lambda_{\vec{n}}^2}{Q^2\Lambda^2}\right), \tag{41}$$

where $s(x) = x\, n_{\mathrm{F}}(x) + \ln(1 + e^{-x})$ is the fermionic entropy function with $n_{\mathrm{F}}(x) = (e^x + 1)^{-1}$. The normalized Fisher metric is extracted via finite differencing:

$$g_{QQ}^{\mathrm{num}} = \frac{S_{\mathrm{vN}}(Q+\delta) - 2S_{\mathrm{vN}}(Q) + S_{\mathrm{vN}}(Q-\delta)}{\delta^2\, S_{\mathrm{vN}}(1)}\,, \tag{42}$$

with $\delta = 10^{-4}$ and evaluated at $Q = 1$.

For the equal-period torus with $\Lambda L = 20$ (ensuring $> 10^4$ modes within the cutoff), we obtain [NEW]:

$$g_{QQ}^{\mathrm{num}}(1)\big|_{d=4} = 8.000 \pm 0.002 = d\times(d-2)(d-3)/d = 4\times 2\,. \tag{43}$$

Since all $d = 4$ dimensions are rescaled simultaneously under constant conformal rescaling, $g_{QQ}/d = 2.000 \pm 0.001$, confirming Eq. (27) to better than 0.02%.

We also compute g_{QQ}^{num} on T^d for $d = 2, 3, 4, 5, 6$. Table III confirms agreement with $(d-2)(d-3)$ in every case.

VIII. WHAT DOESN'T WORK

We state the limitations of the present construction explicitly.

(i) $F^2 \neq |\nabla\Sigma_{\mathrm{EM}}|^2$.—The gauge field strength $F^a_{\mu\nu}F^{a\,\mu\nu}$ is the squared norm of a 2-form, *not* of the gradient of a scalar. There exists no scalar field Σ_{EM} such that $F_{\mu\nu}F^{\mu\nu} = |\nabla\Sigma_{\mathrm{EM}}|^2$ on a general 4-manifold. The reason is topological: a 2-form F can have non-trivial cohomology class $[F] \in H^2(M;\mathbb{R})$, which obstructs writing $F = d\Sigma_{\mathrm{EM}}$ (that would require F to be exact, hence cohomologically trivial). The gauge sector is Fisher information of a *1-form*, not a 0-form. This is the fundamental reason why $\Sigma_{\mathrm{grav}} \oplus \Sigma_{\mathrm{gauge}}$ is an orthogonal decomposition, not a unification into a single Σ.

(ii) The internal algebra $\mathcal{A}_F$ is an input.—The finite spectral triple F with algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ that generates the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is not derived from the Σ framework. It is a separate input to the noncommutative geometry [8, 9]. Whether $\mathcal{A}_F$ can be constrained by information-theoretic principles is an open question.

(iii) The cosmological constant term.—The a_0 coefficient ($\propto \mathrm{Vol}(M)$) scales as Q^d and has Fisher metric $g_{QQ}^{(0)} = d(d-1)/Q^2 = 12/Q^2$ in $d = 4$. This does not match $\Sigma = 2\ln Q$; it matches $\Sigma = 12\ln Q$ (if such an identification were made). The cosmological constant does not fit the Fisher framework as a Σ-type quantity.

(iv) Fermion masses.—The internal Dirac operator D_F encodes Yukawa couplings and fermion masses through the finite spectral triple [8]. This structure is entirely separate from the conformal Σ and is not addressed here.

(v) Constant conformal factor.—Our scaling analysis uses spatially uniform Q. For position-dependent $Q(x)$, the a_4 term develops a non-trivial variation (the conformal anomaly), and the simple formula $g_{QQ} = (d-2)(d-3)/Q^2$ acquires gradient corrections [17]. The full result for general $Q(x)$ requires the Chamseddine–Connes dilaton construction [17], which we defer to future work.

(vi) Euclidean vs. Lorentzian signature.—The spectral action and CCSvS entropy are defined in Euclidean signature. Our $\Sigma = 2\ln Q$ is Lorentzian. The Wick rotation is standard for static spacetimes but not rigorously established for the full spectral triple [23, 24].

(vii) The Khronon is not an inner fluctuation.—In the NCG framework, gauge fields arise as *inner* fluctuations of D [7, 11]. The Khronon field enters through *outer* automorphisms (conformal rescaling), not inner fluctuations. The ghost condensation mechanism $K(Q) = \mu^2(Q-1)^2$ [19] is additional structure not part of the standard spectral triple.

IX. DISCUSSION

A. The "why $d = 4$" result

The strongest new contribution of this paper is the derivation of $d = 4$ from $(d-2)(d-3) = 2$. This is an *algebraic* condition—not a dynamical selection, not an anthropic observation, and not a compactification from higher dimensions. It states: the spacetime dimension is the unique value for which the conformal Fisher information of the gravitational action equals the entropy production of the quantum channel that encodes temporal asymmetry.

The argument does not assume specific matter content (it holds for any $\mathcal{A}_F$), specific topology (it holds on any closed manifold), or specific dynamics (it uses only the heat kernel expansion, which is kinematic). The only inputs are: (1) the spectral action principle, (2) the identification $\Sigma = 2\ln Q$ from the Petz recovery framework, and (3) the requirement that these two structures match at the level of Fisher information.

B. Connection to Papers I–VI

Paper I [1] established $\Sigma = -\ln F \geq 0$ as a universal measure of temporal asymmetry. Paper II [2] proved $\Sigma = 2\ln Q$ on static backgrounds via the gravitational channel theorem. Paper III [3] extended this to FRW backgrounds with $Q = 1 + \delta$ and resolved the $c_s^2 = 0$ constraint. Paper VI [4] established the DBI-Σ correspondence for electromagnetism.

The present paper identifies the origin of the coefficient "2" in $\Sigma = 2\ln Q$: it is $(d-2)(d-3)$ evaluated at $d = 4$. It also explains why the gravitational and gauge sectors decouple: they are Fisher information functionals at different form degrees (0-form vs. 1-form), orthogonal under conformal variation.

C. The spectral action as entropy potential

The CCSvS result [12] identifies the von Neumann entropy with the spectral action: $S_{\mathrm{vN}} = \mathrm{Tr}\, h(\beta D)$. Our result identifies Σ with the Fisher information (second variation) of this entropy. The spectral action is therefore the "entropy potential" whose curvature in field space generates the temporal asymmetry measure Σ.

The Einstein–Hilbert term a_2 is the sector whose Fisher curvature matches $\Sigma = 2\ln Q$; the matter sector a_4 is flat (conformally invariant) and does not contribute. The hierarchy a_0, a_2, a_4, $a_6, \ldots$ is a hierarchy of Fisher curvatures:

$$g_{QQ}^{(k)}(1) = (d-2k)(d-2k-1)\big|_{d=4} = \{12,\, 2,\, 0,\, 6,\, \ldots\}\,, \tag{44}$$

of which only $g_{QQ}^{(1)} = 2$ matches Σ.

D. The Dorau–Much connection

Dorau and Much [18] recently derived the semiclassical Einstein equations from QRE on bifurcate Killing horizons. Their result corresponds to the *first-order* variation of Σ (the entropic force $d\Sigma/dQ = 2/Q$), while our Fisher metric is the *second-order* variation. The two are consistent: Einstein's equations arise from $\delta\Sigma = 0$ at first order (equations of motion); the Fisher metric characterizes the information geometry of the solution space at second order (fluctuations around equilibrium).

E. Towards the Standard Model from Σ

The conformal invariance of a_4 in $d = 4$ means that gauge fields and the Higgs potential do not affect Σ at the level of constant conformal variation. The gauge sector enters through *position-dependent* variations (inner fluctuations of D), which generate the Yang–Mills and Higgs equations as Euler–Lagrange conditions of the spectral action [7, 11]. A full unification requires extending the present analysis from constant to general conformal factors, incorporating both inner and outer fluctuations. This is the subject of ongoing work (Papers 7–8).

F. Open questions

Several questions remain:

1. Can the internal algebra $\mathcal{A}_F$ be derived from Σ or from information-theoretic principles?
2. Does the position-dependent extension $Q(x)$ modify the dimension selection, or is $(d-2)(d-3) = 2$ robust?
3. Can fermion masses (encoded in D_F) be connected to the Fisher framework?
4. Is there a Lorentzian spectral triple that makes the Wick rotation unnecessary?

These are non-trivial open problems. The present paper establishes the foundation: the spectral action and the Σ framework are connected through Fisher information, and this connection selects $d = 4$.

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